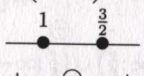
$$\Rightarrow 2\left(x-\frac{3}{2}\right)(x-1) \leq 0$$

$MNB \Rightarrow$ 

$$\mathbf{K} = \left\langle \mathbf{1}; \tfrac{3}{2} \right\rangle$$

3.1.34 :

$$\begin{aligned} -(3+2x)^2 &< 7x+15 \\ -9-12x-4x^2 &< 7x+15 \\ -4x^2-19x-24 &< 0 \qquad /\cdot(-1) \\ 4x^2+19x+24 &> 0 \end{aligned}$$

$D = 361 - 384 = -13$

dosazením např. $x = 0 \Rightarrow 24 > 0\ V_p!$

$$\Rightarrow \quad \mathbf{K} = \mathcal{R}$$

Příklady k procvičení

Řešte v $\mathcal{R}$:

3.1.P.1 :	$x^2 - 36 < 0$	$K = (-6;6)$
3.1.P.2 :	$x^2 - 2 \geq 0$	$K = (-\infty;-\sqrt{2}\rangle \cup \langle\sqrt{2};\infty)$
3.1.P.3 :	$4x^2 - 9 < 0$	$K = (-\frac{3}{2};\frac{3}{2})$
3.1.P.4 :	$2x^2 - 3 \leq 0$	$K = \langle -\sqrt{\frac{3}{2}};\sqrt{\frac{3}{2}}\rangle$
3.1.P.5 :	$x^2 + 5 > 0$	$K = \mathcal{R}$
3.1.P.6 :	$3x^2 + 1 < 0$	$K = \emptyset$
3.1.P.7 :	$x^2 - 2x \leq 0$	$K = \langle 0;2\rangle$
3.1.P.8 :	$2x^2 + 5x > 0$	$K = (-\infty;-\frac{5}{2}) \cup (0;\infty)$
3.1.P.9 :	$x - 3x^2 < 0$	$K = (-\infty;0) \cup (\frac{1}{3};\infty)$
3.1.P.10 :	$3x^2 - 8x \geq 0$	$K = (-\infty;0\rangle \cup \langle\frac{8}{3};\infty)$
3.1.P.11 :	$x^2 + 2x + 1 \leq 0$	$K = \{-1\}$
3.1.P.12 :	$x^2 + 6x + 9 > 0$	$K = \mathcal{R} - \{-3\}$
3.1.P.13 :	$x^2 - 5x + 6 < 0$	$K = (2;3)$
3.1.P.14 :	$x^2 - 5x - 24 > 0$	$K = (-\infty;-3) \cup (8;\infty)$
3.1.P.15 :	$x^2 - 6x + 8 \geq 0$	$K = (-\infty;2\rangle \cup \langle 4;\infty)$
3.1.P.16 :	$x^2 - 7x - 8 > 0$	$K = (-\infty;-1) \cup (8;\infty)$
3.1.P.17 :	$x^2 + 2x - 15 \leq 0$	$K = \langle -5;3\rangle$
3.1.P.18 :	$x^2 + 3x + 2 \leq 0$	$K = (-\infty;-2\rangle \cup \langle 1;\infty)$
3.1.P.19 :	$x^2 + 5x + 7 > 0$	$K = \mathcal{R}$
3.1.P.20 :	$x^2 - 2x + 2 > 0$	$K = \mathcal{R}$
3.1.P.21 :	$x^2 - 2x + 1 < 0$	$K = \emptyset$
3.1.P.22 :	$x^2 + 6x + 10 < 0$	$K = \emptyset$
3.1.P.23 :	$x^2 - 12x + 36 \leq 0$	$K = \{6\}$
3.1.P.24 :	$9x^2 - 6x + 1 \leq 0$	$K = \{\frac{1}{3}\}$

3.1.P.25 :	$-x^2 + 4x - 4 < 0$	$K = \mathcal{R} - \{2\}$
3.1.P.26 :	$3x^2 - 7x + 4 \leq 0$	$K = \langle 1, \frac{4}{3}\rangle$
3.1.P.27 :	$3x^2 + 5x - 2 < 0$	$K = (-2;\frac{1}{3})$
3.1.P.28 :	$-2x^2 - 5x + 12 > 0$	$K = (-4;\frac{3}{2})$
3.1.P.29 :	$x^2 - 2x - 2 \leq 0$	$K = \langle 1-\sqrt{3};1+\sqrt{3}\rangle$
3.1.P.30 :	$3x^2 - 7x + 6 < 0$	$K = \emptyset$
3.1.P.31 :	$2(1-3x)^2 \leq 3x - 1$	$K = \langle\frac{1}{3};\frac{1}{2}\rangle$
3.1.P.32 :	$-(3-2x)^2 < 7x - 15$	$K = (-\infty;-\frac{3}{4}) \cup (2;\infty)$
3.1.P.33 :	$-4(3-x)^2 \geq 11x - 33$	$K = \langle\frac{1}{4};3\rangle$
3.1.P.34 :	$(2-3x) - (x-1) < -4 - (x-2)^2$	$K = (4-\sqrt{5};4+\sqrt{5})$
3.1.P.35 :	$(1-1,5x)(x-1) < -2 - 0,5(x-2)^2$	$K = (-\infty;-\frac{3}{2}) \cup (2;\infty)$
3.1.P.36 :	$(x-2\sqrt{2})^2 \geq 16 - \sqrt{2}x$	$K = (-\infty;-\sqrt{2}\rangle \cup \langle 4\sqrt{2};\infty)$

Řešené příklady

Řešte v $\mathcal{R}$:

3.2.1 :

$$\frac{3}{x-x^2-1} < 0 \qquad /\cdot(-1) \qquad x^2 - x + 1 \neq 0 \qquad \mathcal{D}: \boxed{x \in \mathcal{R}}$$

$$\frac{3}{x^2-x+1} > 0 \qquad \left(\tfrac{+}{\oplus} = +\right)$$

$\Rightarrow x^2 - x + 1 > 0$

$D = 1 - 4 = -3$

dosazením např. $x = 0 \Rightarrow 1 > 0\ V_p!$

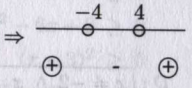
$$\Rightarrow \mathbf{K} = \mathbf{R}$$

3.2.2 :

$$\frac{x^2+9}{x^2-16} \geq 0 \qquad \left(\tfrac{+}{\oplus} = +\right) \qquad x^2 - 16 \neq 0 \qquad \mathcal{D}: \boxed{x \neq \pm 4}$$

$$\Rightarrow \quad x^2 - 16 > 0$$

$$\Rightarrow \quad (x+4)(x-4) > 0$$

$MNB \Rightarrow$ 

$$\mathbf{K} = (-\infty;-\mathbf{4}) \cup (\mathbf{4};\infty)$$

3.2.3 :

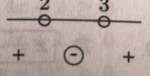
$$\frac{x^2-5x+6}{x^2+x+1} < 0 \qquad x^2 + x + 1 \neq 0 \qquad \mathcal{D}: \boxed{x \in \mathcal{R}}$$

ČIT: $V.v. \Rightarrow \quad x_1 = 2 \ \vee \ x_2 = 3$

JM: $D = 1 - 4 = -3$

dosazením např. $x = 0 \Rightarrow 1 > 0\ V_p! \Rightarrow x^2 + x + 1 > 0$

$\Rightarrow \left(\frac{\oplus}{+} = -\right) \Rightarrow (x-2)(x-3) < 0$

$MNB \Rightarrow$ 

$$\mathbf{K} = (\mathbf{2};\mathbf{3})$$